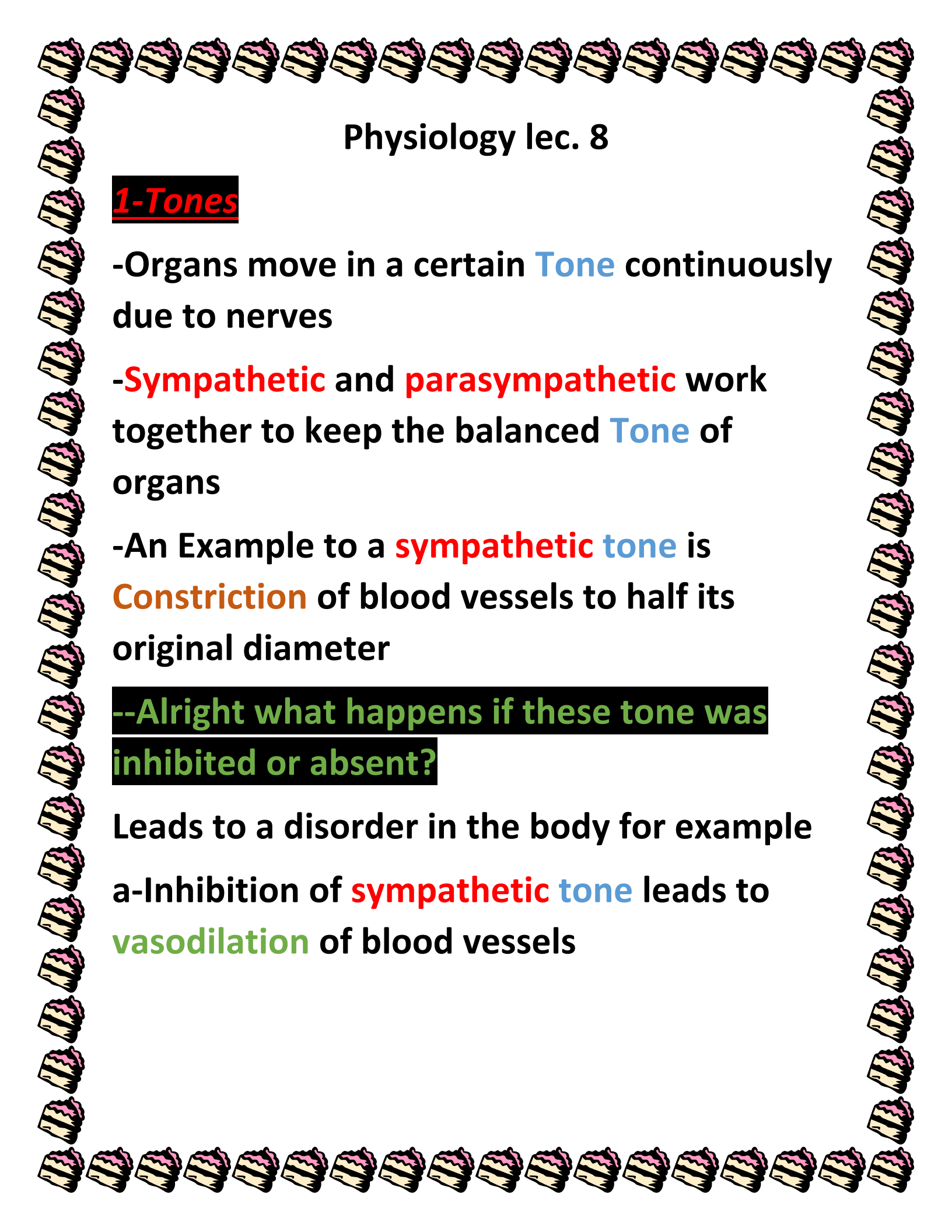




Physiology lec. 8

1-Tones

-Organs move in a certain **Tone** continuously due to nerves

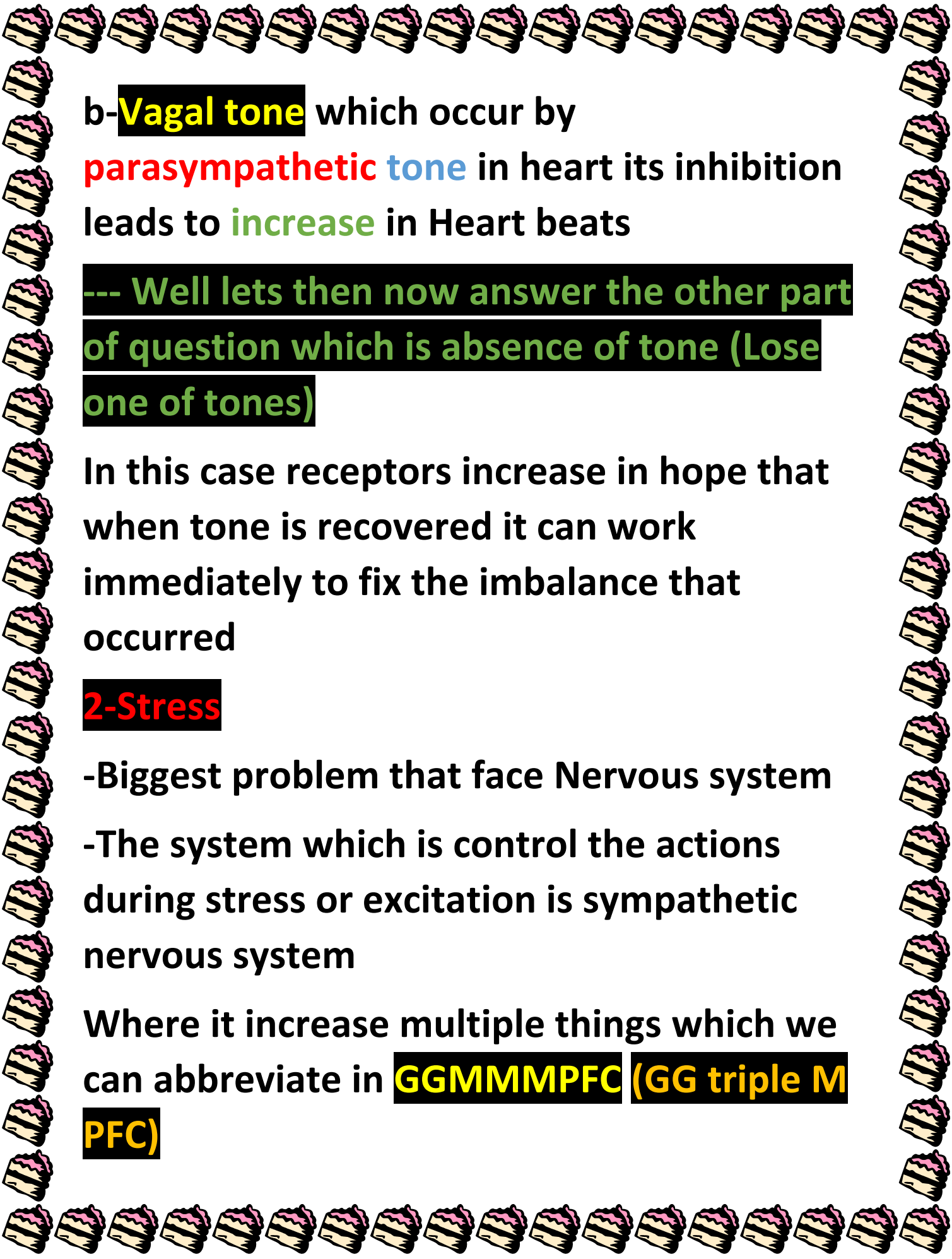
-**Sympathetic** and **parasympathetic** work together to keep the balanced **Tone** of organs

-An Example to a **sympathetic tone** is **Constriction** of blood vessels to half its original diameter

--Alright what happens if these tone was inhibited or absent?

Leads to a disorder in the body for example

a-Inhibition of **sympathetic tone** leads to **vasodilation** of blood vessels



b-**Vagal tone** which occur by
parasympathetic tone in heart its inhibition
leads to **increase** in Heart beats

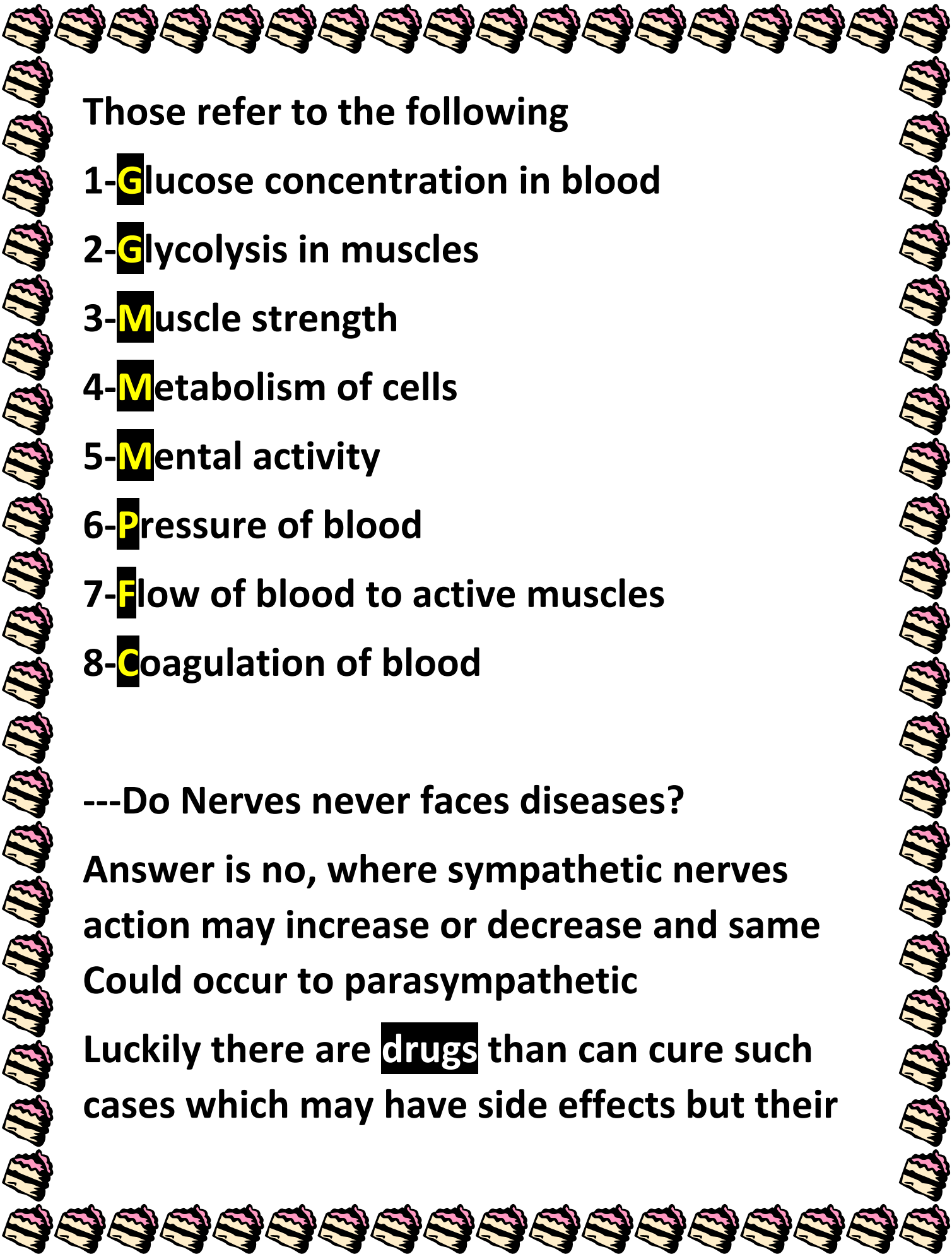
--- Well lets then now answer the other part
of question which is absence of tone (Lose
one of tones)

In this case receptors increase in hope that
when tone is recovered it can work
immediately to fix the imbalance that
occurred

2-Stress

- Biggest problem that face Nervous system
- The system which is control the actions
during stress or excitation is sympathetic
nervous system

Where it increase multiple things which we
can abbreviate in **GGMMMPFC** (**GG triple M
PFC**)



Those refer to the following

1-**G**lucose concentration in blood

2-**G**lycolysis in muscles

3-**M**uscle strength

4-**M**etabolism of cells

5-**M**ental activity

6-**P**ressure of blood

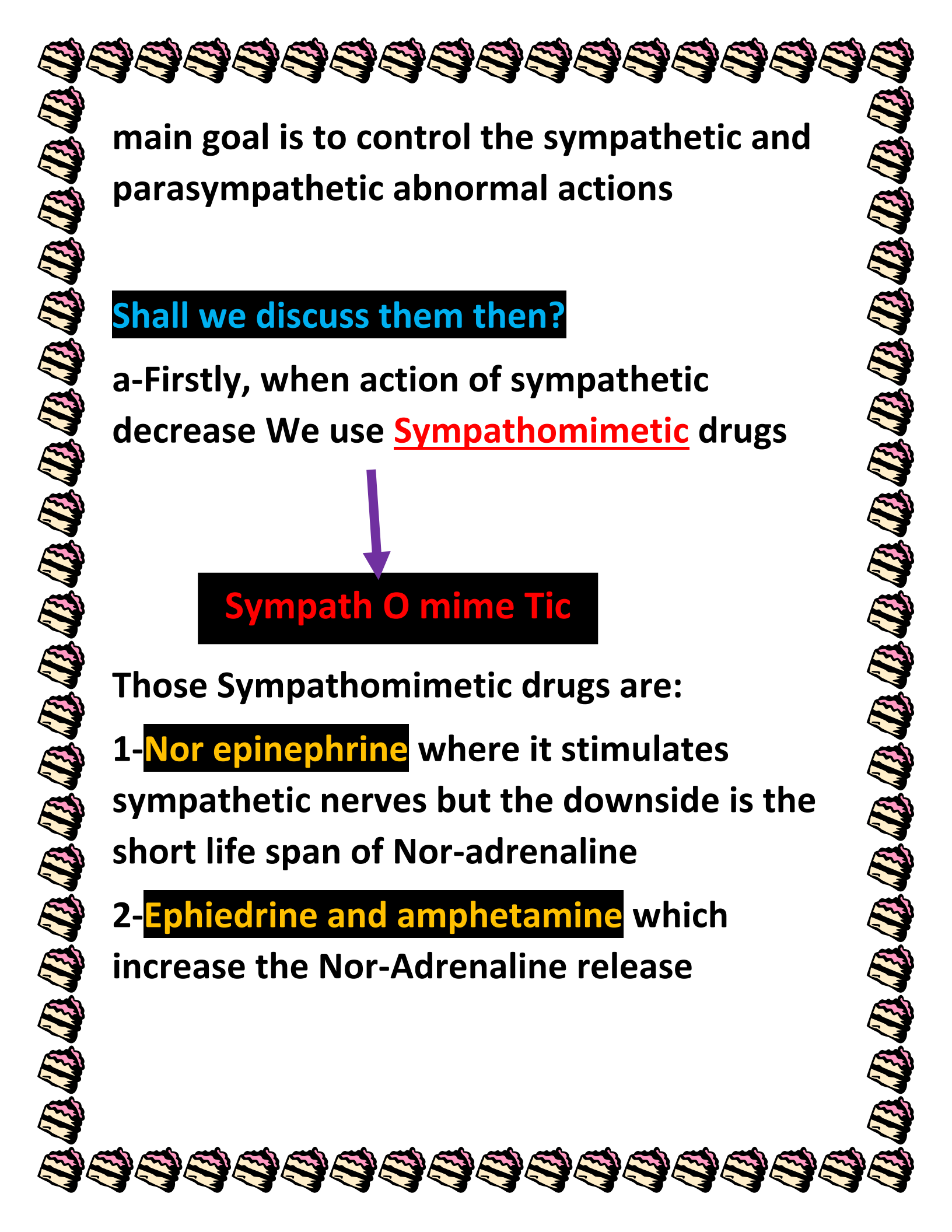
7-**F**low of blood to active muscles

8-**C**oagulation of blood

---Do Nerves never faces diseases?

Answer is no, where sympathetic nerves action may increase or decrease and same Could occur to parasympathetic

Luckily there are **drugs** than can cure such cases which may have side effects but their



main goal is to control the sympathetic and parasympathetic abnormal actions

Shall we discuss them then?

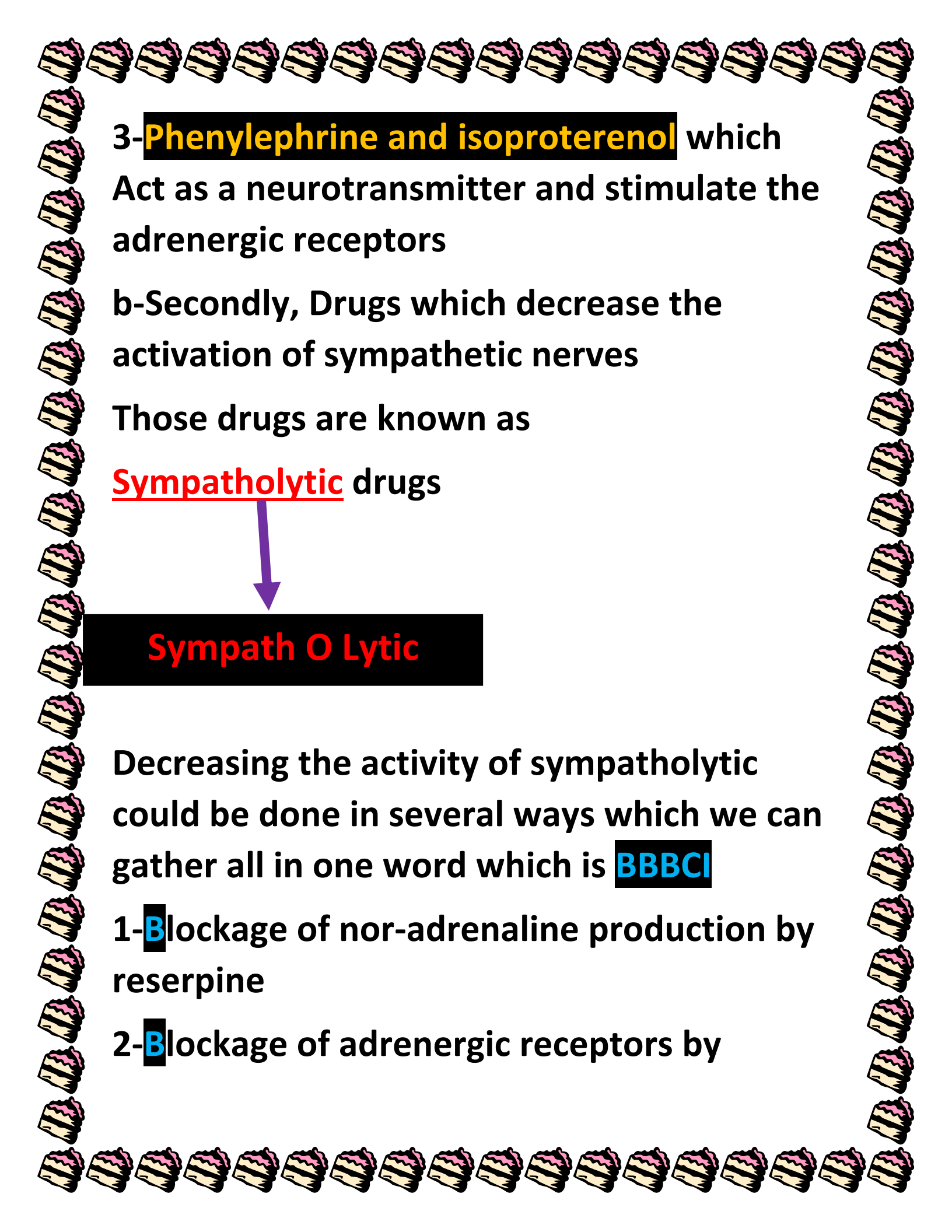
a-Firstly, when action of sympathetic decrease We use Sympathomimetic drugs



Sympath O mime Tic

Those Sympathomimetic drugs are:

- 1-**Nor epinephrine** where it stimulates sympathetic nerves but the downside is the short life span of Nor-adrenaline
- 2-**Ephedrine and amphetamine** which increase the Nor-Adrenaline release



3- **Phenylephrine and isoproterenol** which
Act as a neurotransmitter and stimulate the
adrenergic receptors

b- Secondly, Drugs which decrease the
activation of sympathetic nerves

Those drugs are known as

Sympatholytic drugs

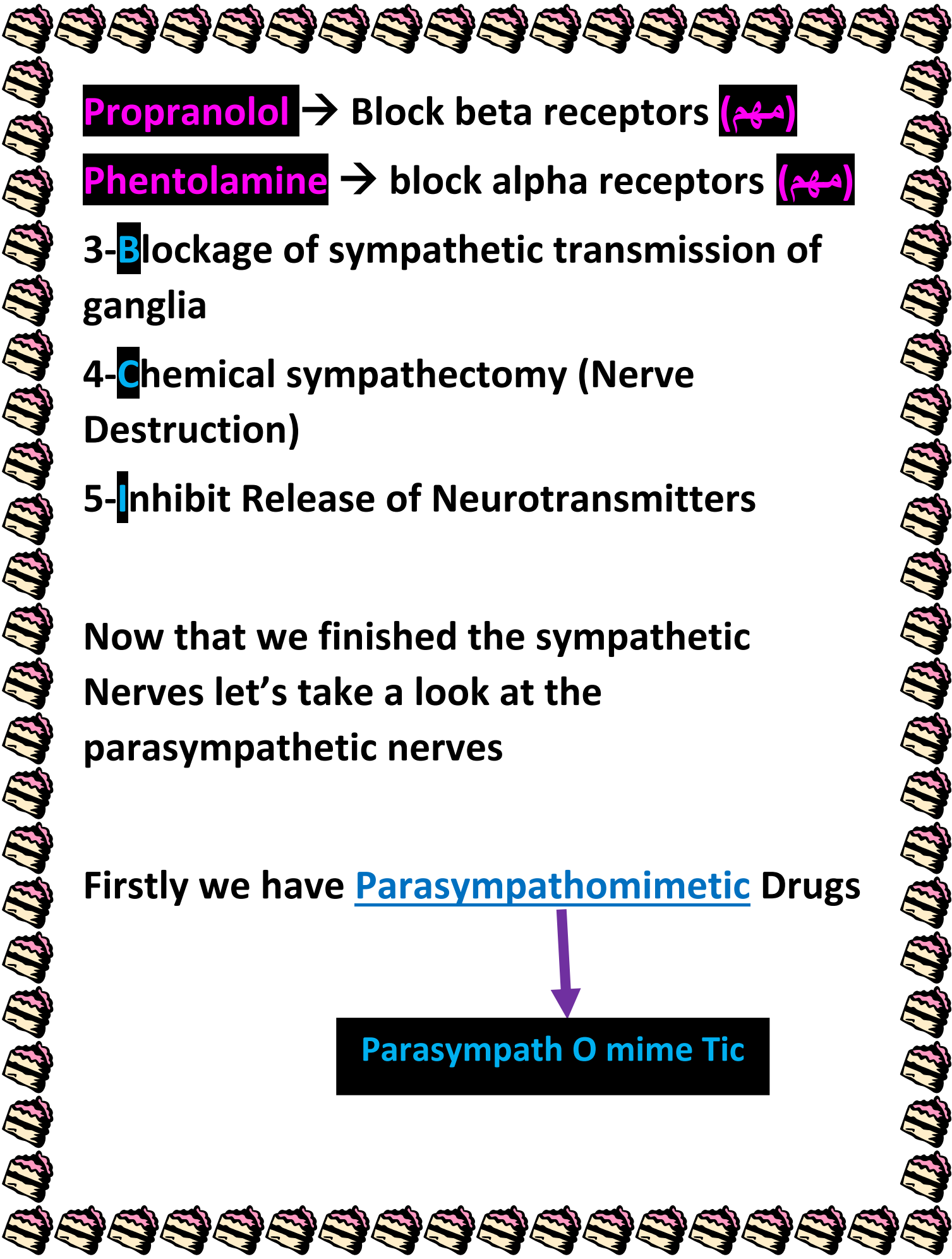


Sympath O Lytic

Decreasing the activity of sympatholytic
could be done in several ways which we can
gather all in one word which is **BBBCI**

1- **B**lockage of nor-adrenaline production by
reserpine

2- **B**lockage of adrenergic receptors by



Propranolol → Block beta receptors (مهم)

Phentolamine → block alpha receptors (مهم)

3-Blockage of sympathetic transmission of ganglia

4-Chemical sympathectomy (Nerve Destruction)

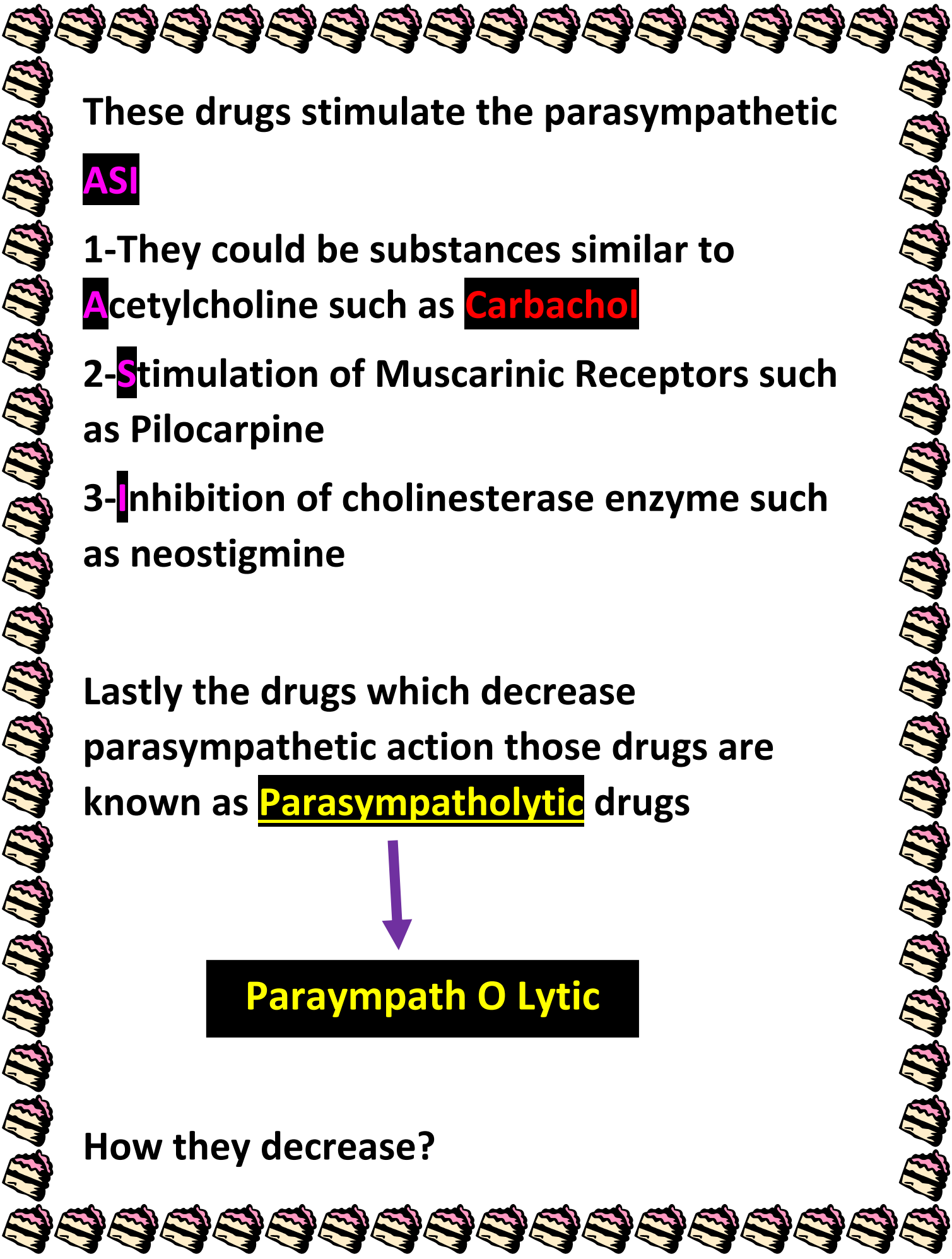
5-Inhibit Release of Neurotransmitters

Now that we finished the sympathetic Nerves let's take a look at the parasympathetic nerves

Firstly we have Parasympathomimetic Drugs



Parasympath O mime Tic



These drugs stimulate the parasympathetic

ASI

1-They could be substances similar to

Acetylcholine such as **Carbachol**

2-**S**timulation of Muscarinic Receptors such as Pilocarpine

3-**I**nhibition of cholinesterase enzyme such as neostigmine

Lastly the drugs which decrease parasympathetic action those drugs are known as **Parasympatholytic** drugs



Parasympatholytic

How they decrease?



Simply, by blockage of cholinergic activity

Those actions are:

BIA

1-Blockage of Cholinergic receptors such as

atropine (حفظ)

2-Inhibition of acetylcholine production and its release

3-Activation of cholinesterase enzyme

❄❄Made By : Karim Maged ❄❄